Menu Driven Case Study: Student Performance Analyzer

Scenario

A training institute wants to build a **menu-driven system** to manage student performance records.

The system should allow the trainer to perform different operations using a menu.

Each student record includes:

- Student Name
- Python Marks
- SQL Marks
- Linux Marks

The program should continue running until the trainer chooses **Exit**.

Input Constraints

- Marks range: **0–100**
- Student name: **string**
- Minimum students: **1**

Menu Options

- Add Student Record
- Display All Student Records
- Search Student by Name
- Show Class Average
- Exit

Grading Criteria

Average Marks	Grade
≥ 85	A

70 – 84	B
50 – 69	C
< 50	Fail

Sample Program Execution

Program Start

```
Student Performance System

1. Add Student
2. Display Students
3. Search Student
4. Class Average
5. Exit

Enter your choice: 1
```

Step 1: Add First Student

```
Enter student name: Ravi
Enter Python marks: 85
Enter SQL marks: 80
Enter Linux marks: 90

Student added successfully
```

Menu Appears Again

```
Student Performance System

1. Add Student
2. Display Students
3. Search Student
4. Class Average
5. Exit

Enter your choice: 1
```

Step 2: Add Second Student

```
Enter student name: Anu
Enter Python marks: 60
Enter SQL marks: 65
Enter Linux marks: 70

Student added successfully
```

Menu Appears Again

```
Student Performance System

1. Add Student
2. Display Students
3. Search Student
4. Class Average
5. Exit

Enter your choice: 2
```

Step 3: Display All Students

```
Name: Ravi
Marks: [85, 80, 90]
Total: 255
Average: 85.0
Grade: A

Name: Anu
Marks: [60, 65, 70]
Total: 195
Average: 65.0
Grade: C
```

Menu Appears Again

```
Student Performance System

1. Add Student
2. Display Students
3. Search Student
4. Class Average
5. Exit
```

```
Enter your choice: 3
```

Step 4: Search Student

```
Enter student name to search: Ravi

Student Found
Name: Ravi
Marks: [85, 80, 90]
Total: 255
Average: 85.0
Grade: A
```

Menu Appears Again

```
Student Performance System

1. Add Student
2. Display Students
3. Search Student
4. Class Average
5. Exit

Enter your choice: 4
```

Step 5: Class Average

```
Class Average: 75.0
```

Calculation:

```
Ravi Average = 85
Anu Average = 65

Class Average = (85 + 65) / 2 = 75
```

Exit Program

```
Student Performance System

1. Add Student
2. Display Students
```

```
3. Search Student
4. Class Average
5. Exit

Enter your choice: 5

Exiting program...
```